

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Medium

Question

The length of one side of square M is **6** times the length of one side of square N. The area of square N is **49** square feet. What is the area, in square feet, of square M?

Answer

- A. **1,764**
- B. 504
- C. 294
- D. 252

Correct Answer: A

Rationale

Choice A is correct. The area, ***A***, of a square is given by the equation **$A = s^2$** , where ***s*** is the length of one side of the square. Let ***x*** represent the length, in feet, of one side of square N. It's given that the area of square N is **49** square feet; therefore, it follows that **$49 = x^2$** . It's also given that the length of one side of square M is **6** times the length of one side of square N. Therefore, the length of one side of square M can be represented by the expression **$6x$** . Substituting **$6x$** for ***s*** in the equation **$A = s^2$** yields **$A = (6x)^2$** , or **$A = 36x^2$** . Since **$49 = x^2$** , substituting **49** for **x^2** in the equation **$A = 36x^2$** yields **$A = 36(49)$** , or **$A = 1,764$** . Therefore, the area, in square feet, of square M is **1,764**.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.